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Title: Movie watchlist Manager using Doubly linked list in c++

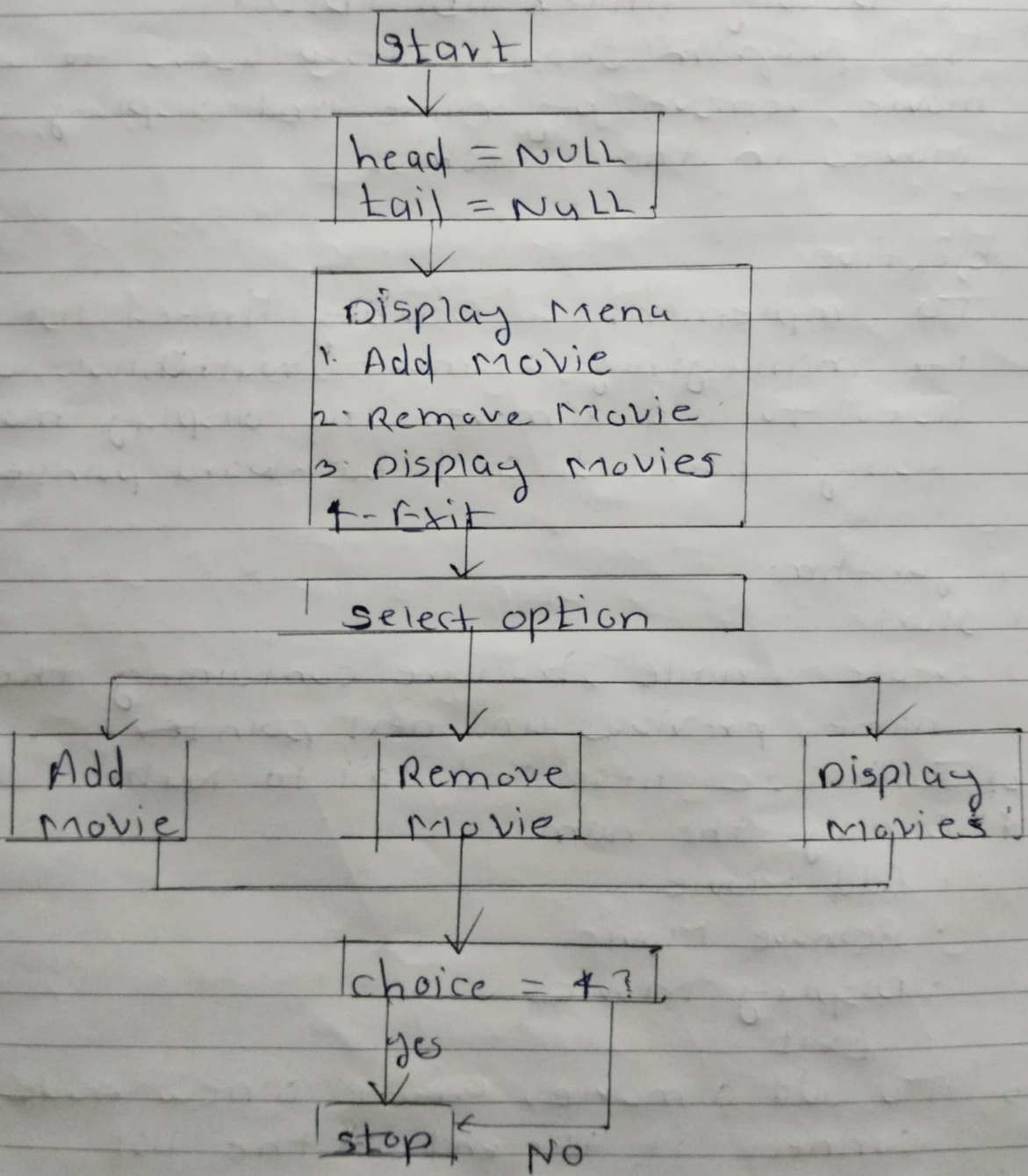
problem statement:
create a Movie watchlist Manager using a Doubly linked list (DLL).
The program should allow users to add movie, remove by name and display all Movies in the watchlist.

objective:
To implement a Doubly linked list in c++ for managing a movie watchlist, allowing user to add, remove, and display movies using forward and backward pointers

Algorithm:
1. start
2. create a node structure containing the movie name, previous and next pointer.
3. Initialize head and tail to Null.
4. Display the name.
   - Add Movie.
   - Remove Movie
   - Display Movies
   - Exit
5. To add a movie, create a new node and connect it the end of the list.
6. To remove a movie, search by name and update the previous and next pointe.
7. To display movies, traverse the list from head to tail.

8. Repeat until the user selects Exit.
9. Stop

Flowchart:



```
#include <iostream>
#include <string>
using namespace std;

struct Node
{
   string movie;
   Node* prev;
   Node* next;
};

// Add movie
void addMovie(Node*& head, string name)
{
   Node* newNode = new Node;
   newNode->movie = name;
   newNode->prev = NULL;
   newNode->next = NULL;

   if (head == NULL)
   {
      head = newNode;
   }
   else
   {
     Node* temp = head;

     while (temp->next != NULL)
```

```
    {
        cout << "Movie not found.\n";
        return;
    }

    if (temp->prev != NULL)
        temp->prev->next = temp->next;
    else
        head = temp->next;

    if (temp->next != NULL)
        temp->next->prev = temp->prev;

    delete temp;
    cout << "Movie removed successfully.\n";
}

// Display movies
void displayMovies(Node* head)
{
    if (head == NULL)
    {
        cout << "Watchlist is empty.\n";
        return;
    }

    Node* temp = head;
```

```
    cout << "\nMovie Watchlist:\n";

    while (temp != NULL)
    {
        cout << temp->movie << " <-> ";
        temp = temp->next;
    }

    cout << "NULL\n";
}

int main()
{
    Node* head = NULL;
    int choice;
    string name;

    do
    {
        cout << "\n--- Movie Watchlist Manager ---\n";
        cout << "1. Add Movie\n";
        cout << "2. Remove Movie\n";
        cout << "3. Display Movies\n";
        cout << "4. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        switch (choice)    {
```

```
        case 1:
            cout << "Enter movie name: ";
            cin.ignore();
            getline(cin, name);
            addMovie(head, name);
            break;

        case 2:
            cout << "Enter movie name to remove: ";
            cin.ignore();
            getline(cin, name);
            removeMovie(head, name);
            break;

        case 3:
            displayMovies(head);
            break;

        case 4:
            cout << "Program ended.\n";
            break;

        default:
            cout << "Invalid choice.\n";
    }

} while (choice != 4)
return 0;}
```

```
--- Movie Watchlist Manager ---
1. Add Movie
2. Remove Movie
3. Display Movies
4. Exit
Enter choice: 1

Enter movie name: Avengers
Movie added successfully.

--- Movie Watchlist Manager ---
1. Add Movie
2. Remove Movie
3. Display Movies
4. Exit
Enter choice: 1

Enter movie name: Interstellar
Movie added successfully.

--- Movie Watchlist Manager ---
1. Add Movie
2. Remove Movie
3. Display Movies
4. Exit
Enter choice: 1

Enter movie name: Inception
Movie added successfully.

--- Movie Watchlist Manager ---
1. Add Movie
2. Remove Movie
3. Display Movies
4. Exit
Enter choice: 3

Movie Watchlist:
Avengers -> Interstellar ->
Inception -> NULL

--- Movie Watchlist Manager ---
1. Add Movie
2. Remove Movie
3. Display Movies
4. Exit
Enter choice: 2

Enter movie name to remove:
Interstellar
Movie removed successfully.

--- Movie Watchlist Manager ---
1. Add Movie
2. Remove Movie
3. Display Movies
4. Exit
Enter choice: 3

Movie Watchlist:
Avengers -> Inception -> NULL

--- Movie Watchlist Manager ---
1. Add Movie
2. Remove Movie
3. Display Movies
4. Exit
Enter choice: 4

Program ended.
```